Boundary-Specific Uncertainty in Transformer-Based Named Entity Recognition:

FROM BOUNDARY-AWARE TO UNCERTAINTY-ADAPTIVE CONFORMAL SCORING

BY UMRAH

EXAMPLE

The BARACK OBAMA administration visited London

How certain is the model about
Where an entity begins and ends ?

NER must decide both what the entity is and where it begins and Ends

EXAMPLE

The **BARACK OBAMA** administration visited London

START

END

CORRECT SPAN

Barak Obama

TOO SHORT

Barak

TOO LONG

Barak Obama administration

One Prediction needs TWO decisions

TYPE DECISION

What is the entity ?

PERSON. ORG. LOC. MISC

BOUNDARY DECISION

Where does it START and END

Even a correct type can become wrong entity

THIS PROJECT FOCUSES ON UNCERTAINTY AT THE START AND END BOUNDARIES

RESEARCH GAP

Boundary-aware NER, uncertainty estimation and conformal prediction already exist - but their combination leaves a specific question underexplored.

Boundary-aware NER

Uncertainty estimation

GAP

A direct test of whether boundary-specific uncertainty can be quantified, then progressively integrated into typed-span conformal scoring.

Conformal prediction

RESEARCH AIM AND Objectives

Quantify start- and end-boundary uncertainty in Transformer-based NER and evaluate what happens when it is incorporated into typed-span conformal scoring.

01.

Objective

Construct start- and end-boundary uncertainty for predicted entities.

02.

Objective

Test whether boundary uncertainty identifies incorrect boundaries.

03.

Objective

Integrate it into typed-span conformal scoring through M1 - M4.

04.

Objective

Evaluate robustness under controlled text corruption and reproduce the methodology in the WNUT-17 setting.

OVERALL PIPELINE

1 NER Models

Two Transformer-based architectures

BERT (BIO)

Barack	Obama	visited	Paris
B-PER	I-PER	O	B-LOC

Sequence labelling

Span model (SpanNER-style)

Barack	Obama	visited	Paris

Span representation

2 Boundary-Specific Uncertainty

Architecture-specific start and end uncertainty

PER

Barack	Obama	visited	Paris

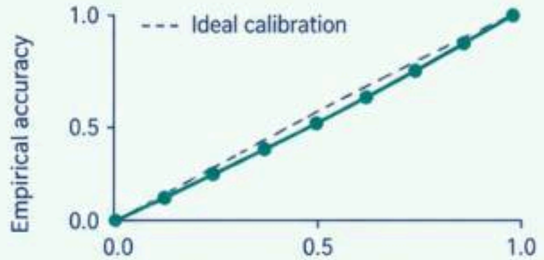
Start uncertainty **End uncertainty**

Estimated separately for the start and end boundaries of each candidate span

Core contribution

3 Probability Calibration

Calibrate model probabilities using temperature scaling



Schematic reliability diagram

Temperature scaling

Rescales logits without changing the predicted class

4 Conformal Candidate Features

Recomputed from temperature-scaled probabilities

PER

Barack	Obama	visited	Paris

Typed-span confidence c

$R_s = 0.20$ $R_e = 0.80$ Boundary ranks R_s, R_e

Boundary uncertainty u_s, u_e

5 Progressive Conformal Scoring (M1–M4)

Compare four nonconformity score functions

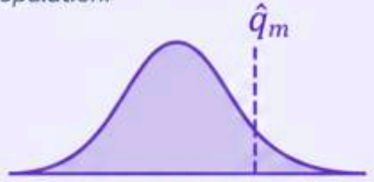
M1	Confidence only
M2	Shared boundary-rank regularisation
M3	Separate start/end boundary-rank regularisation
M4	Uncertainty-adaptive boundary regularisation

Core contribution

6 Independent Conformal Calibration

Method-specific threshold selection

Method-specific q_m estimated from the reserved conformal-calibration population.



7 Prediction Sets (Conformal Output)

Retain candidates with nonconformity scores at or below the calibrated threshold

- (Barack Obama, PER)
- (Obama, PER)
- (Barack, PER)

...

Set size varies with candidate scores and the calibrated threshold.

Two architectures, Two views of a boundary

BERT BIO

Token sequence labelling

Boundaries are represented indirectly through BIO states.

SPAN MODEL

Span classification

Start and end coordinates are explicit candidate properties.

Same concept: local boundary ambiguity. Different evidence.



BERT (BIO) — Beginning/Continuation Evidence

Illustrative proposed span: Barack Obama (PER)

	[CLS]	Barack	Obama	visited	Paris	.	[SEP]
Example sentence (tokens)							
Model BIO tag (predicted)	-	B-PER	I-PER	O	B-LOC	O	-

1 Start Boundary (at proposed start token)

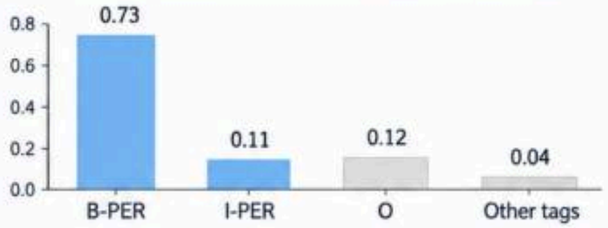
Context position

Token: Barack

Position: s

Predicted tag: B-PER

Model probabilities at position s (token "Barack")



Start uncertainty

$$u_s = \frac{P_s(I\text{-}PER)}{P_s(B\text{-}PER) + P_s(I\text{-}PER) + \epsilon}$$

$$= \frac{0.11}{0.73 + 0.11} \approx 0.13$$

2 End Boundary (at token after the proposed end)

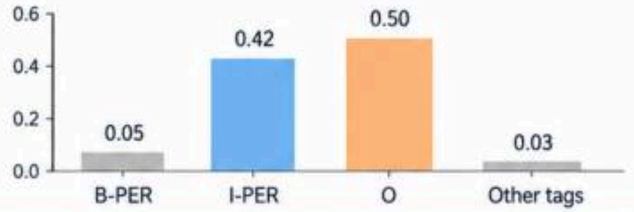
Context position

Token: visited

Position: $e + 1$

Predicted tag: O

Model probabilities at position $e + 1$ (token "visited")



End uncertainty

$$u_e = P_{e+1}(I\text{-}PER) = 0.42$$

If the candidate ends at the final token in the sentence, $u_e = 0$.

Outputs used in later conformal scoring

Span confidence (not shown here)	Start uncertainty	End uncertainty
c	$u_s \approx 0.13$	$u_e = 0.42$

BERT BIO

Token sequence labelling

Boundaries are represented indirectly through BIO states.

BERT BIO

Token sequence labelling

Boundaries are represented indirectly through BIO states.

- Start uncertainty = "**Start here**" vs "**Continue from before**"
- Compare B-type and I-type probability for the same entity class.
- **Higher** I evidence → **less** certain this is the true start.

BERT (BIO) — Beginning/Continuation Evidence

Illustrative proposed span: Barack Obama (PER)

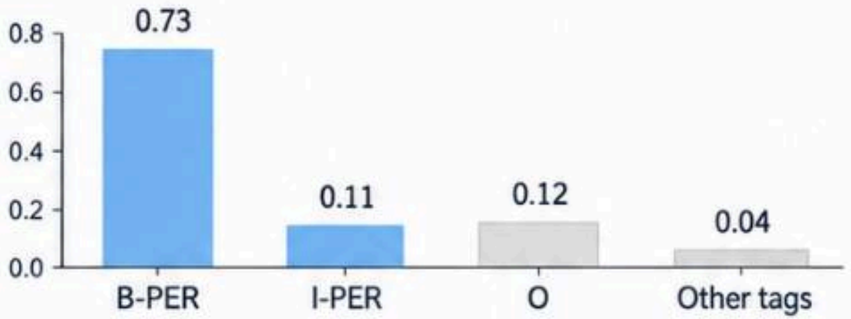
Example sentence (tokens)	[CLS]	Barack	Obama	visited	Paris	.	[SEP]
Model BIO tag (predicted)	-	B-PER	I-PER	O	B-LOC	O	-

1 Start Boundary (at proposed start token)

Context position

Token	Barack
Position	s
Predicted tag	**B-PER**

Model probabilities at position s (token "Barack")



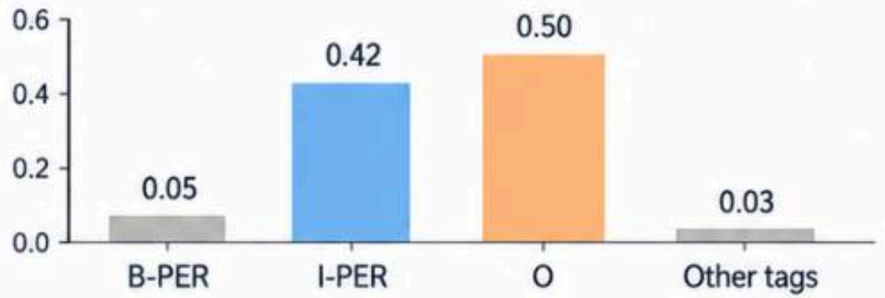
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$$= \frac{0.11}{0.73 + 0.11} \approx \mathbf{0.13}$$

2 End Boundary (at token after the proposed end)

Context position	
Token	visited
Position	$e + 1$
Predicted tag	O



End uncertainty

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BERT BIO

Token sequence labelling

Boundaries are represented indirectly through BIO states.

- End uncertainty = "Should the entity continue?"
- Look at next token's I-type probability
- **Higher** I after end → **higher** end uncertainty

SPAN MODEL

Span classification

Start and end coordinates are explicit candidate properties.

Input sentence (tokenized) — Illustrative proposed span: **Barack Obama (PER)**

The	former	president	Barack	Obama	visited	Paris	on	Monday	.
0	1	2	3	4	5	6	7	8	9

1 Alternative-start competitors (same end $e = 4$, same type PER; vary start s)

Tracked typed-span candidate

Span [3, 4]
"Barack Obama"
Type: PER

Model probability $p_y = \mathbf{0.76}$

Valid alternative starts (same end and type)

Span [2, 4]
"president Barack Obama"
Type: PER
$p_s^* = \mathbf{0.42}$

Span [1, 4]
"former president Barack Obama"
Type: PER
0.28

Other valid alternatives (if any)
...

Start uncertainty

$$u_s = \frac{p_s^*}{p_y + p_s^* + \epsilon} = \frac{0.42}{0.76 + 0.42} \approx \mathbf{0.36}$$

2 Alternative-end competitors (same start $s = 3$, same type PER; vary end e)

Tracked typed-span candidate

Span [3, 4]
"Barack Obama"
Type: PER

Model probability $p_y = \mathbf{0.76}$

Valid alternative ends (same start and type)

Span [3, 5]
"Barack Obama visited"
Type: PER
$p_e^* = \mathbf{0.35}$

Span [3, 6]
"Barack Obama visited Paris"
Type: PER
0.18

Other valid alternatives (if any)
...

End uncertainty

$$u_e = \frac{p_e^*}{p_y + p_e^* + \epsilon} = \frac{0.35}{0.76 + 0.35} \approx \mathbf{0.32}$$

Quantities used in later conformal scoring

Typed-span confidence c	Start uncertainty $u_s \approx \mathbf{0.36}$	End uncertainty $u_e \approx \mathbf{0.32}$

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SPAN MODEL

Span classification

Start and end coordinates are explicit candidate properties.

Was boundary-specific uncertainty informative?

Boundary-specific uncertainty was informative across both architecture-specific constructions.
But informative uncertainty does not automatically imply a better downstream conformal decision rule.

START BOUNDARY

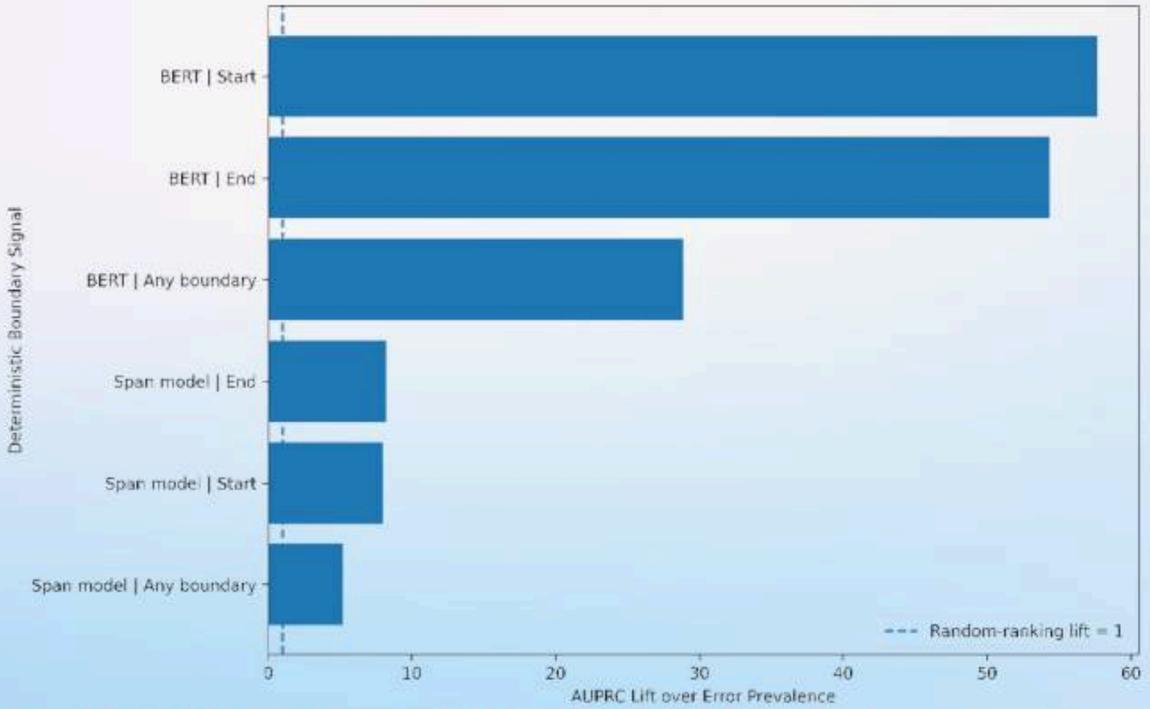
Higher uncertainty identified incorrect starts

BERT-derived signal	Span-derived signal
AUROC	AUROC
0.854	0.853

END BOUNDARY

Higher uncertainty identified incorrect ends

BERT-derived signal	Span-derived signal
AUROC	AUROC
0.938	0.849

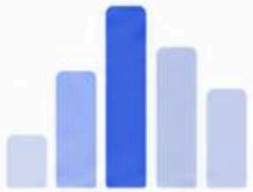


Note: Span AUPRC is lower largely because boundary errors were much rarer in its evaluable population; relative to its own prevalence baseline, both Span signals still achieved about 8× lift.

Before conformal scoring: probability calibration

Temperature scaling corrected probability sharpness before confidence and uncertainty were reused downstream

Saved logits



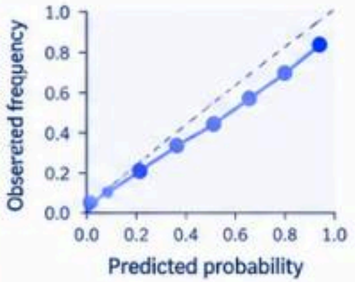
Model output scores (before softmax)

Temperature scaling

$$p_T(y \mid x) = \text{softmax}\left(\frac{z}{T}\right)$$

Soften / sharpen probabilities with T

Calibrated probabilities



Better probability fit

Recompute features



Candidate confidence c + boundary uncertainty u_s, u_e

Calibration changed probability sharpness - not the predicted classes.
The same predefined confidence and boundary-uncertainty constructions were then recomputed from calibrated probabilities.

Saved logits



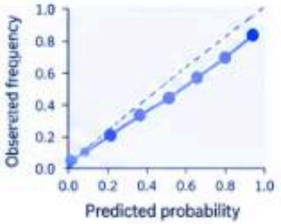
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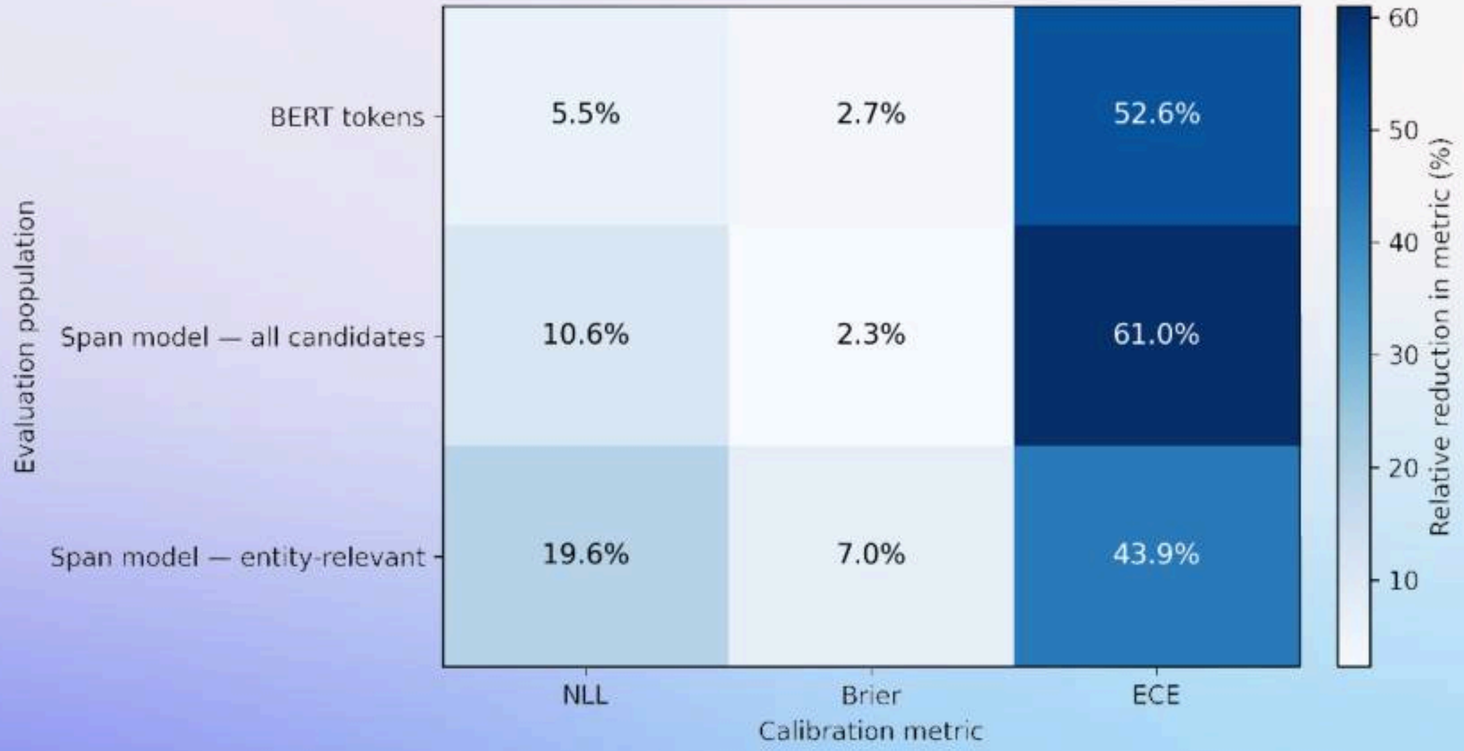


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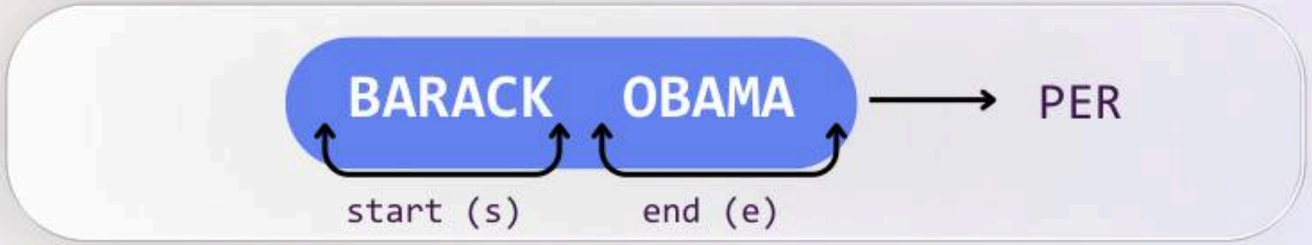
Fitted temperatures

BERT BIO:	$T = 1.264$
Span model:	$T = 1.467$

Temperature scaling improved NLL, Brier and ECE for both architectures without changing the predicted classes.

What enters the conformal score?

Each typed-span candidate is represented by confidence, boundary ranks, and architecture-specific boundary uncertainties.



1 Confidence (c)

How plausible is this typed span?

$$c = P(y \mid x)$$

Typed-span confidence

2 Start rank (R_s)

Where does its start rank among alternative starts?

Candidate start (end, type fixed)	Conf.
1. Barack Obama	0.82
2. Obama	0.46
3. The Barack Obama	0.12

$$R_s = (r_s - 1) \,/\, (m_s - 1)$$

Normalised from 0 to 1

3 End rank (R_e)

Where does its end rank among alternative ends?

Candidate end (start, type fixed)	Conf.
1. Barack Obama	0.82
2. Barack	0.28
3. Barack Obama administration	0.20

$$R_e = (r_e - 1) \,/\, (m_e - 1)$$

Normalised from 0 to 1

4 Start uncertainty (u_s)

Architecture-specific start-boundary ambiguity

BERT BIO:
derived from BIO token-state evidence

Span model:
derived from strongest same-type alternative start

Imported from earlier model-specific construction

5 End uncertainty (u_e)

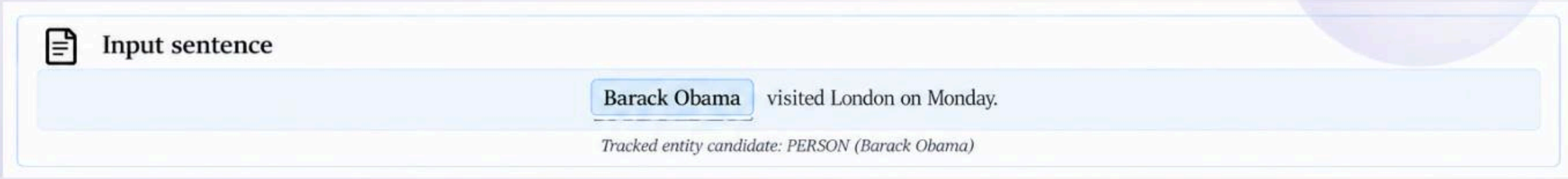
Architecture-specific end-boundary ambiguity

BERT BIO:
derived from continuation evidence after the proposed end

Span model:
derived from strongest same-type alternative end

Imported from earlier model-specific construction

How boundary information entered conformal scoring



Input sentence

Barack Obama visited London on Monday.

Tracked entity candidate: PERSON (Barack Obama)

Tracked candidate (y)

(Barack Obama, PER)

Typed span under evaluation

Candidate inputs

Typed-span confidence (c)	0.74
Start boundary rank (R_s)	0.20
End boundary rank (R_e)	0.80
Start boundary uncertainty (u_s)	0.15
End boundary uncertainty (u_e)	0.70

Method parameters (illustrative)

λ (M2)	0.20
λ_s (M3–M5)	0.50
λ_e (M3–M5)	0.30

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M1
Confidence-only

Nonconformity score

$$a_1 = 1 - c$$

Score (a_1)	**0.26**
Threshold (q_1)	0.20 *(illustrative)*
Decision	Exclude $(a_1 > q_1)$

Illustrative prediction-set excerpt

{ (Obama, PER)
(Barack, PER)
(London, LOC)
(Barack Obama, PER) not included

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M2
Shared boundary-rank

Nonconformity score

$$a_2 = (1 - c) + \frac{1}{2}\lambda(R_s + R_e)$$

Score (a_2)	**0.36**
Threshold (q_2)	0.30 *(illustrative)*
Decision	Exclude ($a_2 > q_2$)

Illustrative prediction-set excerpt

{ (Obama, PER)
(Michelle Obama, PER)
(London, LOC)
(Barack Obama, PER) not included

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Illustrative prediction-set excerpt

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(Michelle Obama, PER)
(London, LOC)
(Barack Obama, PER) not included

M3
Separate start/end boundary-rank

Nonconformity score

$$a_3 = (1 - c) + \frac{1}{2}(\lambda_s R_s + \lambda_e R_e)$$

Score (a_3)	**0.43**
Threshold (q_3)	0.45 *(illustrative)*
Decision	Include ($a_3 \leq \dot{q}_3$)

Illustrative prediction-set excerpt

{ (Barack Obama, PER)
(Obama, PER)
(London, LOC)
(Barack, PER)

Input sentence

Barack Obama visited London on Monday.

Tracked entity candidate: PERSON (Barack Obama)

Tracked candidate (y)

(Barack Obama, PER)

Typed span under evaluation

Candidate inputs

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λ_e (M3–M5)	0.30

M1
Confidence-only

Nonconformity score

$$a_1 = 1 - c$$

Score (a_1)	**0.26**
Threshold (q_1)	0.20 *(illustrative)*
Decision	✗ Exclude ($a_1 > \dot{q}_1$)

Illustrative prediction-set excerpt

{ (Obama, PER)
(Barack, PER)
(London, LOC)
(Barack Obama, PER) not included

M2
Shared boundary-rank

Nonconformity score

$$a_2 = (1 - c) + \frac{1}{2}\lambda(R_s + R_e)$$

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Illustrative prediction-set excerpt

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Separate start/end boundary-rank

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Score (a_3)	**0.43**
Threshold (q_3)	0.45 *(illustrative)*
Decision	✓ Include ($a_3 \leq \dot{q}_3$)

Illustrative prediction-set excerpt

{ (Barack Obama, PER)
(Obama, PER)
(London, LOC)
(Barack, PER)

M4
Uncertainty-adaptive

Nonconformity score

$$a_4 = (1 - c) + \frac{1}{2}[\lambda_s(1 - u_s)R_s + \lambda_e(1 - u_e)R_e]$$

Score (a_4)	**0.34**
Threshold (q_4)	0.28 *(illustrative)*
Decision	✗ Exclude ($a_4 > \dot{q}_4$)

Illustrative prediction-set excerpt

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(Barack Obama, PER) not included

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M1 Confidence-only

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Decision	✗ Exclude ($a_1 > q_1$)

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M2 Shared boundary-rank

Nonconformity score

$$a_2 = (1 - c) + \frac{1}{2}\lambda(R_s + R_e)$$

Score (a_2)	**0.36**
Threshold (q_2)	0.30 *(illustrative)*
Decision	✗ Exclude ($a_2 > q_2$)

Illustrative prediction-set excerpt

{ (Obama, PER)
(Michelle Obama, PER)
(London, LOC)
(Barack Obama, PER) not included

M3 Separate start/end boundary-rank

Nonconformity score

$$a_3 = (1 - c) + \frac{1}{2}(\lambda_s R_s + \lambda_e R_e)$$

Score (a_3)	**0.43**
Threshold (q_3)	0.45 *(illustrative)*
Decision	✓ Include ($a_3 \leq q_3$)

Illustrative prediction-set excerpt

{ (Barack Obama, PER)
(Obama, PER)
(London, LOC)
(Barack, PER)

M4 Uncertainty-adaptive

Nonconformity score

$$a_4 = (1 - c) + \frac{1}{2}[\lambda_s(1 - u_s)R_s + \lambda_e(1 - u_e)R_e]$$

Score (a_4)	**0.34**
Threshold (q_4)	0.28 *(illustrative)*
Decision	✗ Exclude ($a_4 > q_4$)

Illustrative prediction-set excerpt

{ (Obama, PER)
(Barack, PER)
(London, LOC)
(Barack Obama, PER) not included

M5 Confidence-gated

Nonconformity score

$$a_5 = (1 - c) + \frac{1}{2}[\lambda_s(1 - cu_s)R_s + \lambda_e(1 - cu_e)R_e]$$

Score (a_5)	**0.36**
Threshold (q_5)	0.40 *(illustrative)*
Decision	✓ Include ($a_5 \leq q_5$)

Illustrative prediction-set excerpt

{ (Barack Obama, PER)
(Obama, PER)
(Michelle Obama, PER)
(London, LOC)

What happened when boundary uncertainty entered conformal scoring?

Primary nominal 90% conformal results on the locked CoNLL-2003 test set

BERT BIO

Method	Coverage	Mean typed set size
M1	84.74%	2.1911
M2	84.71%	2.1908
M3	84.71%	2.1908
M4	84.71%	2.1908

Under the selected configuration, adding boundary information changed almost nothing.

SPAN MODEL

Method	Coverage	Mean typed set size
M1	84.56%	1.5077
M2/M3	85.09%	1.5138
M3	84.65%	1.5080

M4 changed the trade-off rather than clearly improving it.



Boundary uncertainty was informative, but informative uncertainty did not automatically translate into better conformal prediction sets.

Why did nominal 90% not transfer to 90% test coverage?

EXPLORATORY M5

Confidence gating did not consistently rescue M4.

- Post-freeze, validation-only diagnostic
- BERT: essentially unchanged
- Span: limited benefit at 95%, but no consistent improvement overall
- At 99%, behaviour deteriorated markedly

Why did nominal 90% not transfer to 90% test coverage?

Evaluation of calibration-to-test behaviour on the locked CoNLL-2003 test set

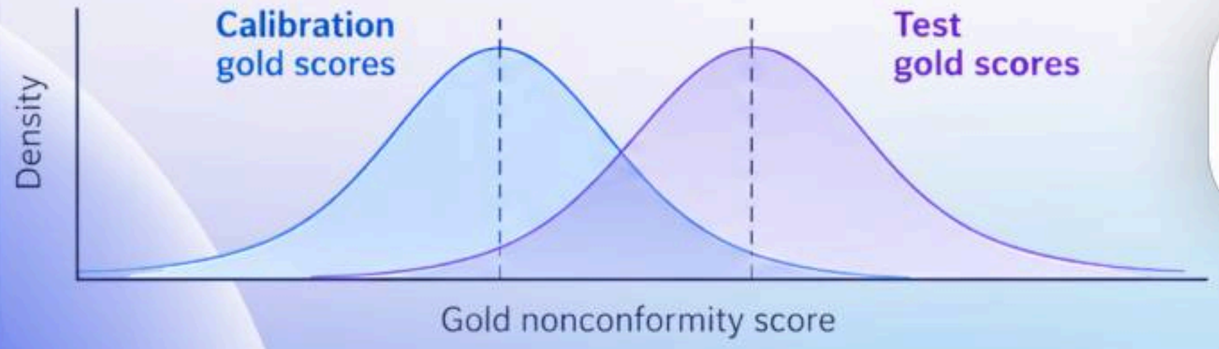
CONFORMAL-CALIBRATION POPULATION

90.15%

apply frozen thresholds to official test set

OFFICIAL CONLL-2003 TEST SET

84.6%– 85.1%



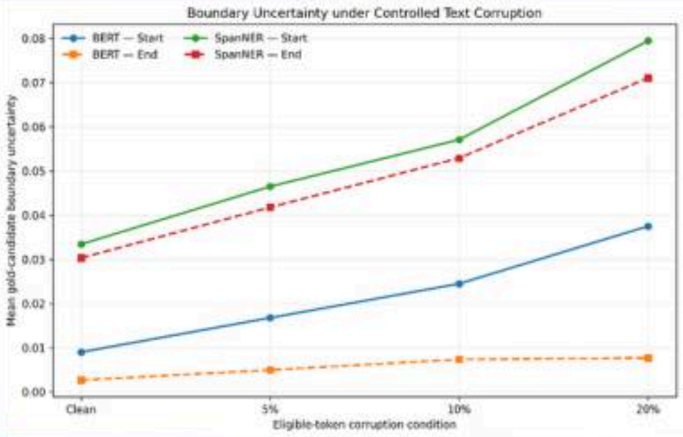
The test set was not identical to the calibration set.
It had a different entity mix and some length differences, and the true entities received higher nonconformity scores overall.

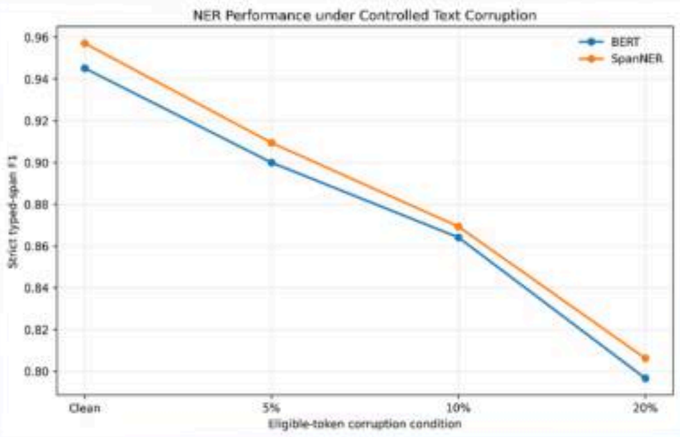
This helps explain the undercoverage, but does not prove one single cause.

Meeting the nominal target on the calibration split did not guarantee the same coverage on the final test set.
This became an important part of the evaluation: uncertainty integration must be judged not only by its design, but by how it behaves when calibration and test score distributions differ.

How did uncertainty behave under Harder Conditions?

CONTROLLED TEXT CORRUPTION



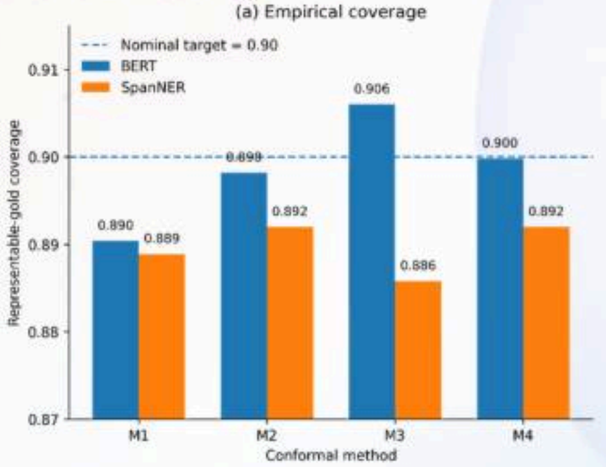


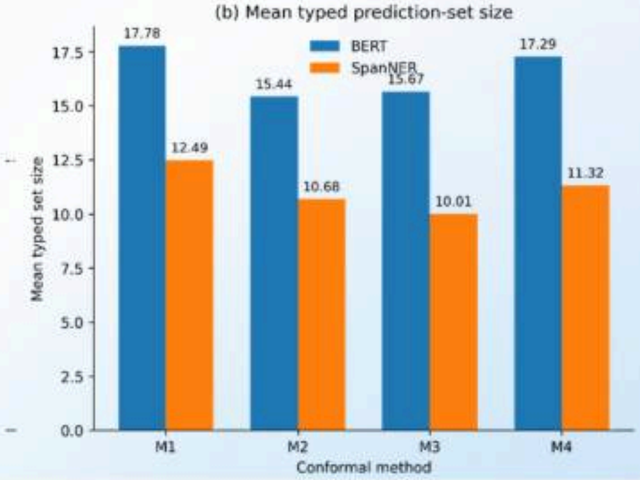
The uncertainty signal moved in the expected direction as recognition became harder.

Robustness testing showed two things: the uncertainty signal responded sensibly to increasing difficulty, while conformal efficiency became more challenging under harder data.

WNUT-17 EXTERNAL SENSITIVITY

The broader methodology was independently reproduced in a harder external NER setting.





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04. Confidence gating did not fully solve it

The exploratory confidence-gated variant changed how uncertainty entered the score, but did not produce a consistent improvement.

Conclusion and Future Work

BOUNDARY-SPECIFIC UNCERTAINTY WAS INFORMATIVE, BUT INFORMATIVE UNCERTAINTY DID NOT AUTOMATICALLY PRODUCE BETTER ADAPTIVE CONFORMAL PREDICTION SETS.



Main contribution

Estimating useful boundary uncertainty and using it effectively are two separate design problems.

01.

Better ambiguity modelling

Distinguish genuine near-ties from weak candidates dominated by stronger alternatives.

02.

Better adaptive integration

Combine confidence, uncertainty, competitor margin and boundary rank more effectively.

03.

Shift-aware conformal calibration

Improve robustness when calibration and test distributions differ.

THANK YOU

